

Below is a **clean, production-style PostgreSQL database design** for your:

# Student Performance & Academic Analytics Platform

This schema is designed to support:

- Role-based authentication
- Student/instructor/admin dashboards
- Courses, enrollment, attendance, grades
- GPA + analytics
- Reports generation
- ML prediction engine (stored outputs)
- Recommendation engine (stored outputs)

It is **normalized, scalable, and analytics-ready** without overengineering.

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## 1. users

### Purpose

Central authentication table for all roles.

| Column        | Type         | Description                  |
|---------------|--------------|------------------------------|
| id            | UUID (PK)    | Unique user ID               |
| email         | VARCHAR(255) | Login email (unique)         |
| password_hash | TEXT         | Hashed password              |
| role          | ENUM         | student / instructor / admin |
| is_active     | BOOLEAN      | Account status               |
| created_at    | TIMESTAMP    | Account creation             |

updated\_at

TIMESTAMP

Last update

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## 2. students

### Purpose

Stores student-specific information.

| Column           | Type                 | Description        |
|------------------|----------------------|--------------------|
| id               | UUID (PK)            | Student ID         |
| user_id          | UUID (FK → users.id) | Linked account     |
| student_id       | VARCHAR(50)          | University roll/ID |
| department_id    | UUID (FK)            | Department         |
| batch_year       | INT                  | Admission year     |
| current_semester | INT                  | Current semester   |
| phone            | VARCHAR(20)          | Optional           |
| profile_photo    | TEXT                 | URL                |
| created_at       | TIMESTAMP            |                    |
| updated_at       | TIMESTAMP            |                    |

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## 3. instructors

### Purpose

Instructor-specific profile.

| Column | Type |
|--------|------|
|--------|------|

|               |                      |
|---------------|----------------------|
| id            | UUID (PK)            |
| user_id       | UUID (FK → users.id) |
| employee_id   | VARCHAR(50)          |
| department_id | UUID (FK)            |
| designation   | VARCHAR(100)         |
| phone         | VARCHAR(20)          |
| created_at    | TIMESTAMP            |
| updated_at    | TIMESTAMP            |

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## 4. departments

### Purpose

Academic departments.

| Column     | Type         |
|------------|--------------|
| id         | UUID (PK)    |
| name       | VARCHAR(100) |
| code       | VARCHAR(20)  |
| created_at | TIMESTAMP    |

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## 5. courses

### Purpose

All courses offered.

| Column        | Type         |
|---------------|--------------|
| id            | UUID (PK)    |
| course_code   | VARCHAR(20)  |
| title         | VARCHAR(255) |
| department_id | UUID (FK)    |
| credit_hours  | INT          |
| description   | TEXT         |
| created_at    | TIMESTAMP    |

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## 6. semesters

### Purpose

Academic sessions.

| Column     | Type        |
|------------|-------------|
| id         | UUID (PK)   |
| name       | VARCHAR(50) |
| start_date | DATE        |
| end_date   | DATE        |

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## 7. course\_offerings

### Purpose

A course in a specific semester taught by an instructor.

| Column        | Type        |
|---------------|-------------|
| id            | UUID (PK)   |
| course_id     | UUID (FK)   |
| instructor_id | UUID (FK)   |
| semester_id   | UUID (FK)   |
| section       | VARCHAR(10) |
| created_at    | TIMESTAMP   |

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## 8. enrollments

### Purpose

Links students to course offerings.

| Column             | Type                              |
|--------------------|-----------------------------------|
| id                 | UUID (PK)                         |
| student_id         | UUID (FK)                         |
| course_offering_id | UUID (FK)                         |
| enrolled_at        | TIMESTAMP                         |
| status             | ENUM (active, dropped, completed) |

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## 9. attendance

### Purpose

Daily attendance tracking.

| Column        | Type                         |
|---------------|------------------------------|
| id            | UUID (PK)                    |
| enrollment_id | UUID (FK)                    |
| date          | DATE                         |
| status        | ENUM (present, absent, late) |
| marked_by     | UUID (FK → instructors.id)   |
| created_at    | TIMESTAMP                    |

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## 10. assessments

### Purpose

All grading components (quiz, exam, assignment).

| Column             | Type                                    |
|--------------------|-----------------------------------------|
| id                 | UUID (PK)                               |
| course_offering_id | UUID (FK)                               |
| title              | VARCHAR(255)                            |
| type               | ENUM (quiz, assignment, midterm, final) |
| total_marks        | INT                                     |
| weightage          | FLOAT                                   |
| due_date           | DATE                                    |
| created_at         | TIMESTAMP                               |

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# 11. grades

## Purpose

Student marks for each assessment.

| Column         | Type       |
|----------------|------------|
| id             | UUID (PK)  |
| assessment_id  | UUID (FK)  |
| student_id     | UUID (FK)  |
| marks_obtained | FLOAT      |
| grade_letter   | VARCHAR(5) |
| grade_points   | FLOAT      |
| created_at     | TIMESTAMP  |

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# 12. gpa\_records

## Purpose

Precomputed GPA per student per semester.

| Column      | Type      |
|-------------|-----------|
| id          | UUID (PK) |
| student_id  | UUID (FK) |
| semester_id | UUID (FK) |
| gpa         | FLOAT     |
| cgpa        | FLOAT     |

|               |           |
|---------------|-----------|
| total_credits | INT       |
| calculated_at | TIMESTAMP |

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## 13. ml\_predictions

### Purpose

Stores ML-based risk prediction results.

| Column            | Type                     |
|-------------------|--------------------------|
| id                | UUID (PK)                |
| student_id        | UUID (FK)                |
| semester_id       | UUID (FK)                |
| risk_score        | FLOAT (0–1)              |
| risk_level        | ENUM (low, medium, high) |
| model_version     | VARCHAR(50)              |
| features_snapshot | JSONB                    |
| created_at        | TIMESTAMP                |

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## 14. recommendations

### Purpose

Stores recommendation engine output.

| Column | Type      |
|--------|-----------|
| id     | UUID (PK) |

|            |                             |
|------------|-----------------------------|
| student_id | UUID (FK)                   |
| type       | VARCHAR(50)                 |
| message    | TEXT                        |
| priority   | ENUM (low, medium, high)    |
| source     | ENUM (rule_based, ml_based) |
| created_at | TIMESTAMP                   |

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## 15. reports

### Purpose

Generated academic reports (PDFs etc.).

| Column        | Type                              |
|---------------|-----------------------------------|
| id            | UUID (PK)                         |
| student_id    | UUID (FK, nullable)               |
| instructor_id | UUID (FK, nullable)               |
| semester_id   | UUID (FK)                         |
| report_type   | ENUM (student, class, department) |
| file_url      | TEXT                              |
| generated_at  | TIMESTAMP                         |

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## 16. department\_analytics

### Purpose

Precomputed admin analytics (for fast dashboards).

| Column         | Type      |
|----------------|-----------|
| id             | UUID (PK) |
| department_id  | UUID (FK) |
| semester_id    | UUID (FK) |
| avg_gpa        | FLOAT     |
| pass_rate      | FLOAT     |
| fail_rate      | FLOAT     |
| avg_attendance | FLOAT     |
| computed_at    | TIMESTAMP |

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## 17. audit\_logs (optional but recommended)

### Purpose

Track system changes.

| Column      | Type      |
|-------------|-----------|
| id          | UUID (PK) |
| user_id     | UUID (FK) |
| action      | TEXT      |
| entity_type | TEXT      |
| entity_id   | UUID      |

timestamp      TIMESTAMP

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# Key Design Decisions (Important)

## 1. Separation of concerns

- users = authentication
- students/instructors = profiles

This avoids role confusion.

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## 2. Analytics optimization

Tables like:

- gpa\_records
- department\_analytics
- ml\_predictions

are **precomputed tables**

👉 This avoids slow real-time aggregation queries.

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## 3. JSONB usage (important)

Used in:

- ml\_predictions.features\_snapshot

Reason:

- ML inputs vary over time
  - flexible storage needed
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## 4. Enrollment-centric design

Everything flows through:

students → enrollments → course\_offerings

This makes attendance, grades, and GPA consistent.

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## 5. Assessment-based grading system

Instead of storing "final grade only", we store:

- quizzes
- assignments
- exams

This enables real analytics like:

- difficulty analysis
  - performance trends
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## 6. Scalability readiness

This schema supports:

- multi-semester data
- multiple departments
- multiple instructors per course (can extend later)
- ML predictions per semester
- historical tracking